

DATA WITH BARAA

SQL COMPONENTS

Baraa Khatib Salkini
DATA WITH BARAA
SQL Course | SELECT Query



```sql
-- Retrieve Customers Data
SELECT
    name,
    LOWER(country)

FROM customers

WHERE country = 'Italy'
```

SQL STATEMENT

```sql
-- Retrieve Customers Data

SELECT

    name,

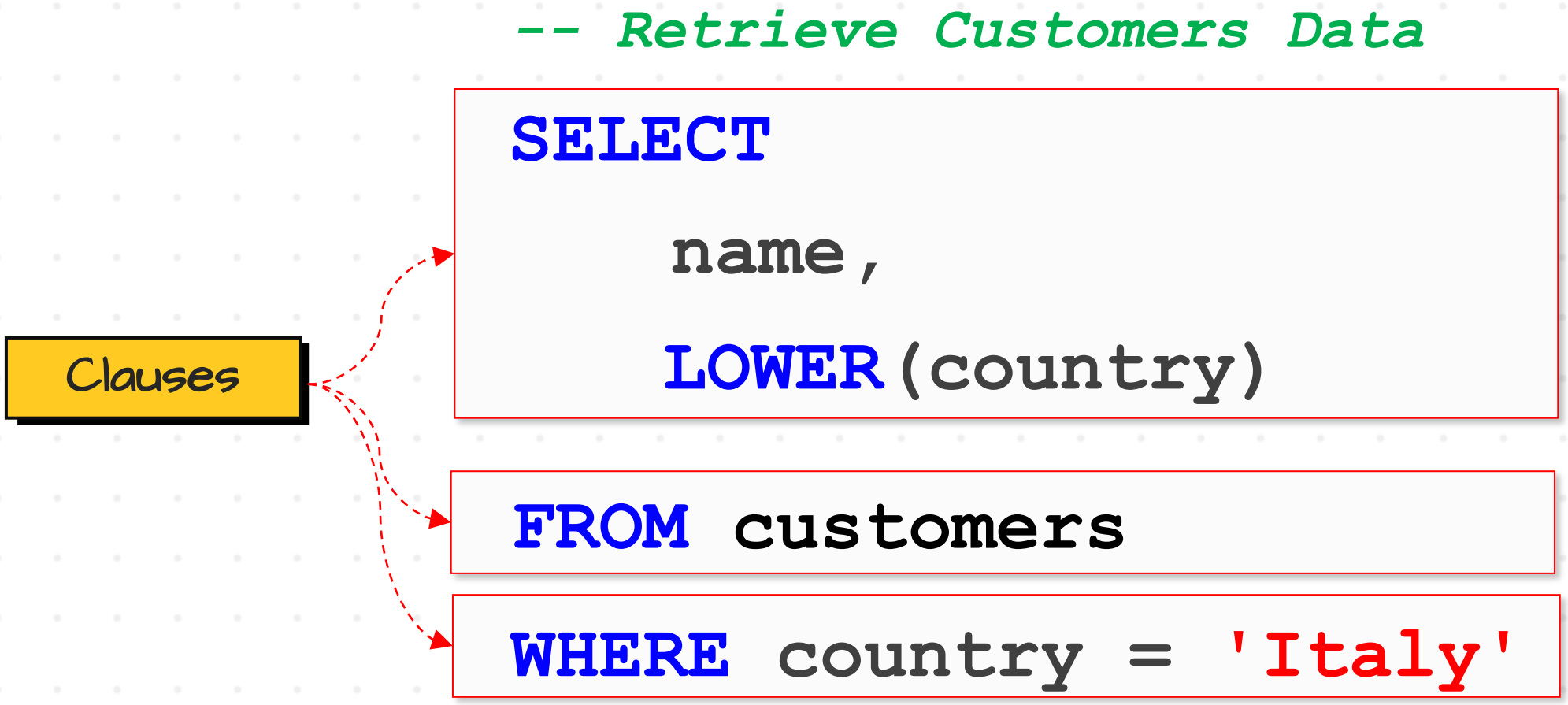
    LOWER(country)

FROM customers

WHERE country = 'Italy'
```

Comment

```sql
-- Retrieve Customers Data
SELECT
    name,
    LOWER(country)
FROM customers
WHERE country = 'Italy'
```


-- Retrieve Customers Data
SELECT
name,
LOWER(country)
FROM customers
WHERE country = 'Italy'
Clauses

```sql
-- Retrieve Customers Data
SELECT
    name,
    LOWER(country)
FROM customers
WHERE country = 'Italy'
```

Keywords

```sql
-- Retrieve Customers Data
SELECT
    name,
    LOWER(country)
FROM customers
WHERE country = 'Italy'
```

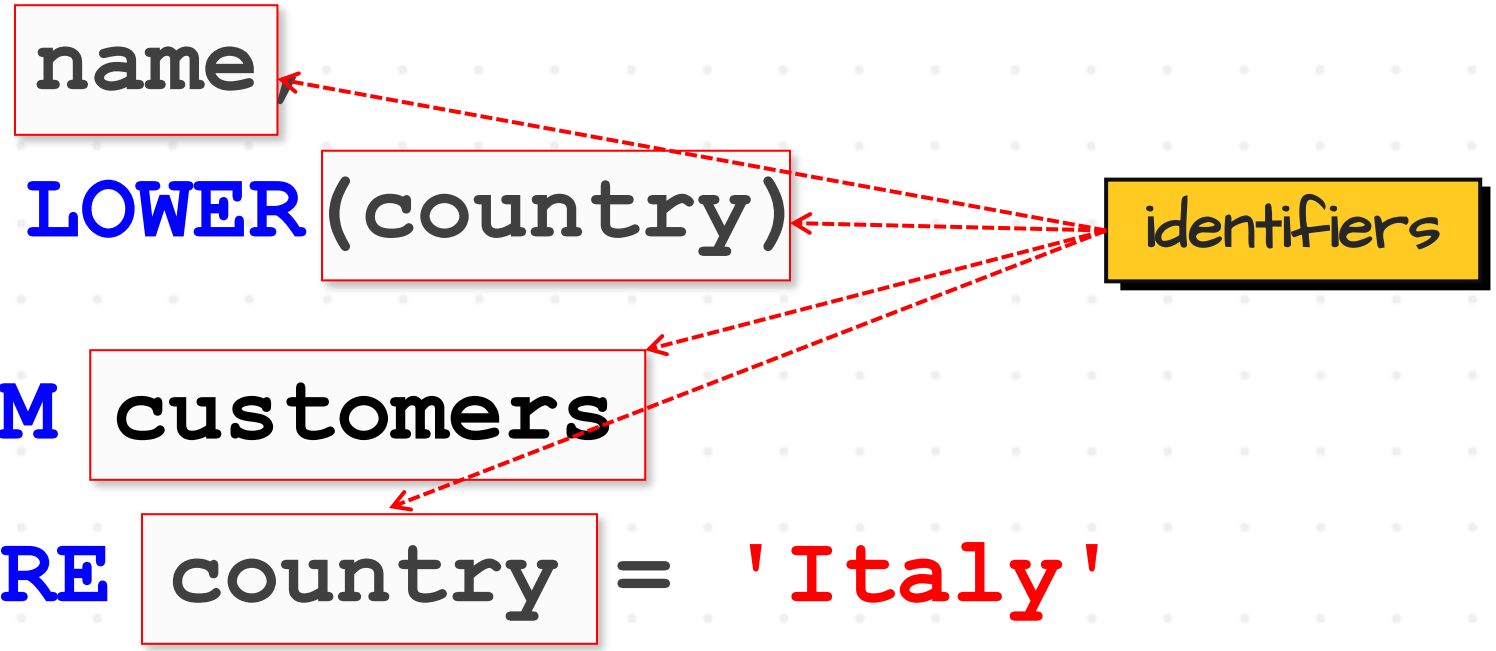
Function

```sql
-- Retrieve Customers Data

SELECT
    name,
    LOWER(country)

FROM customers

WHERE country = 'Italy'
```



```sql
-- Retrieve Customers Data
SELECT
    name,
    LOWER(country)

FROM customers

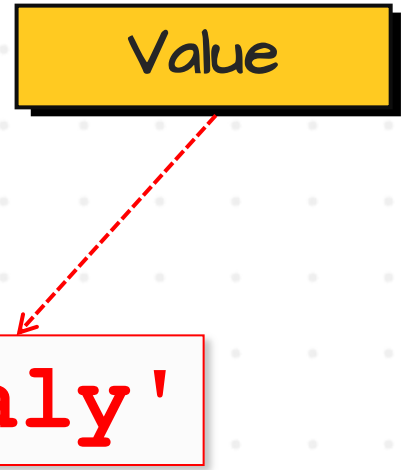
WHERE country = 'Italy'
```



```sql
-- Retrieve Customers Data
SELECT
    name,
    LOWER(country)

FROM customers

WHERE country = 'Italy'
```





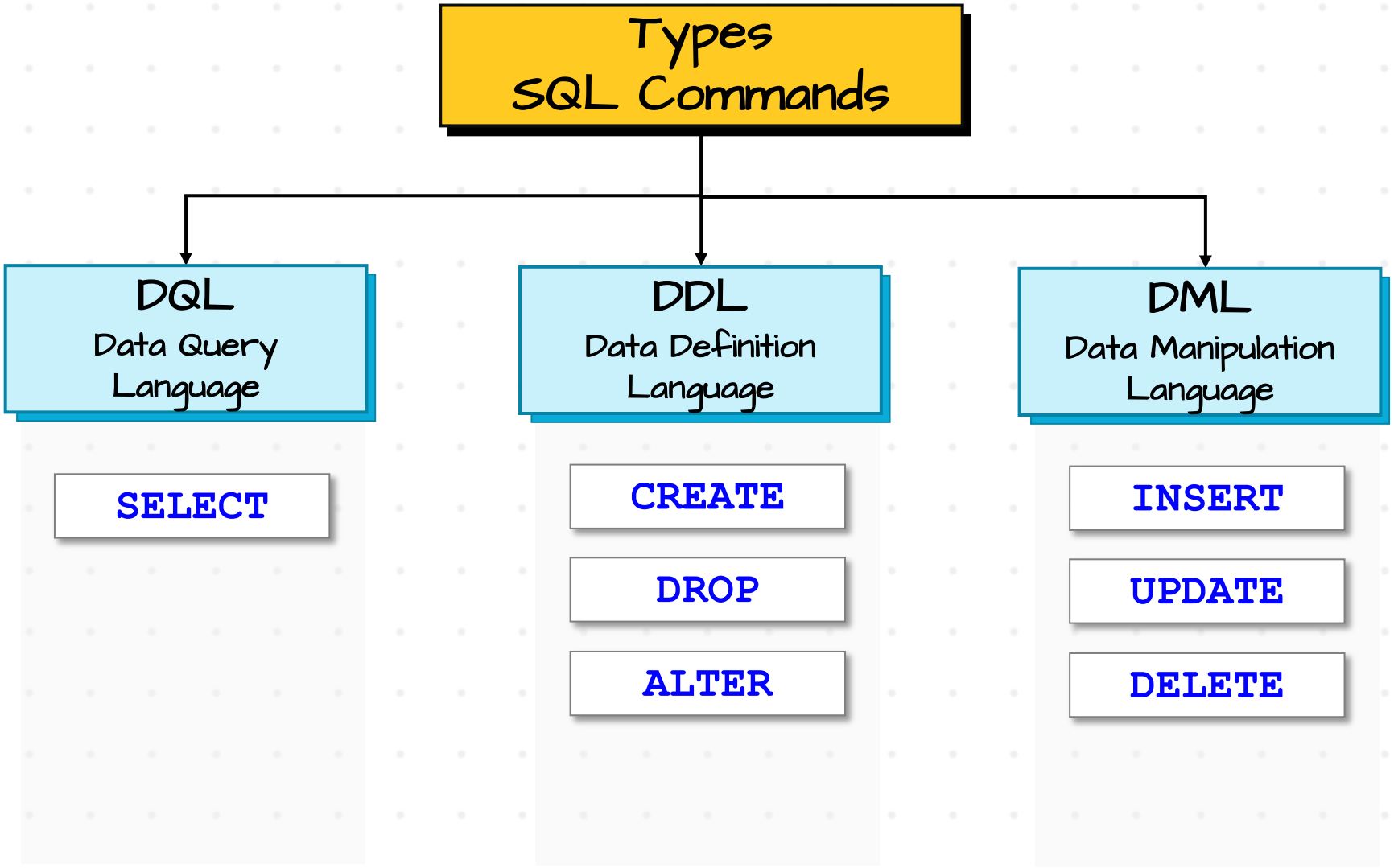
DATA WITH BARAA

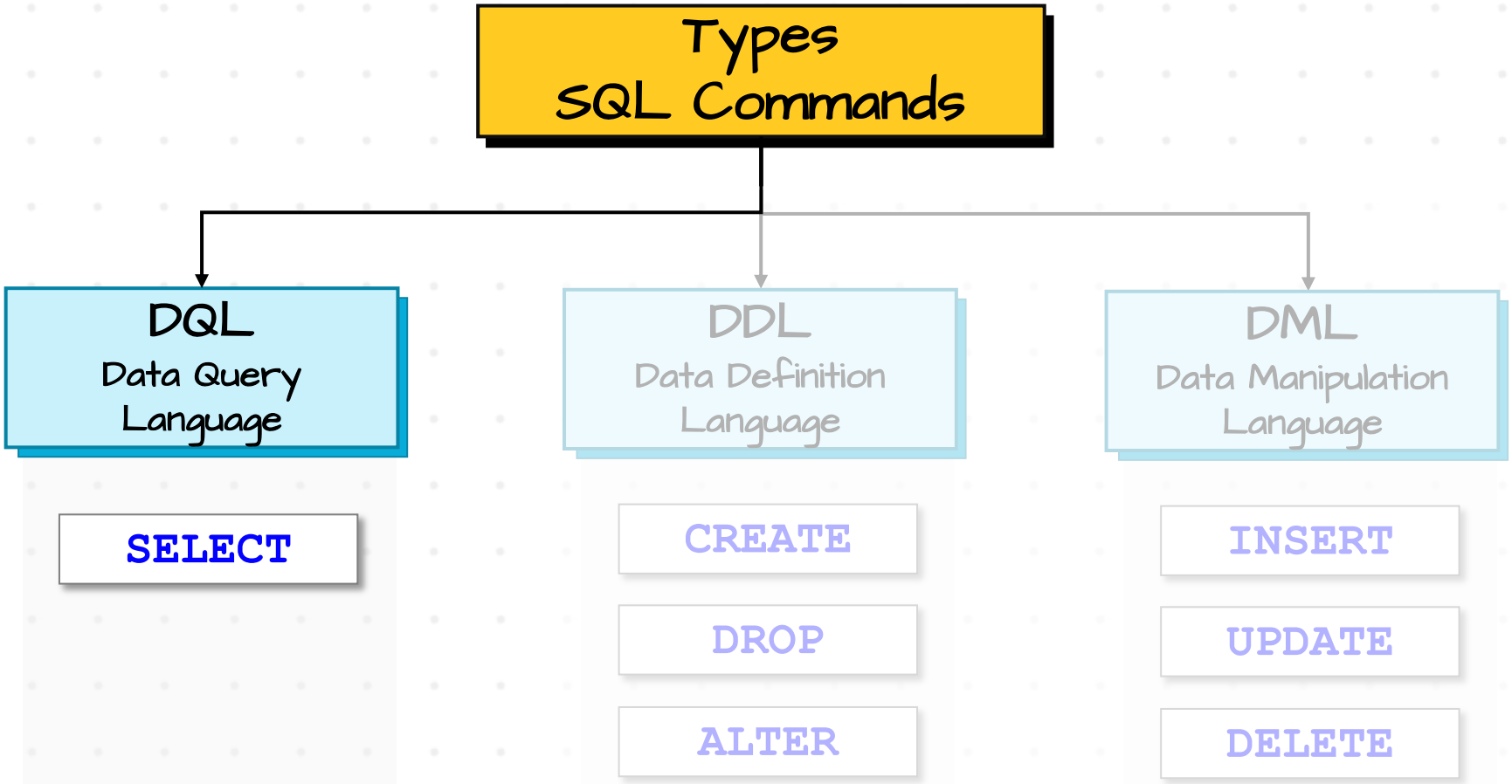
QUERY DATA

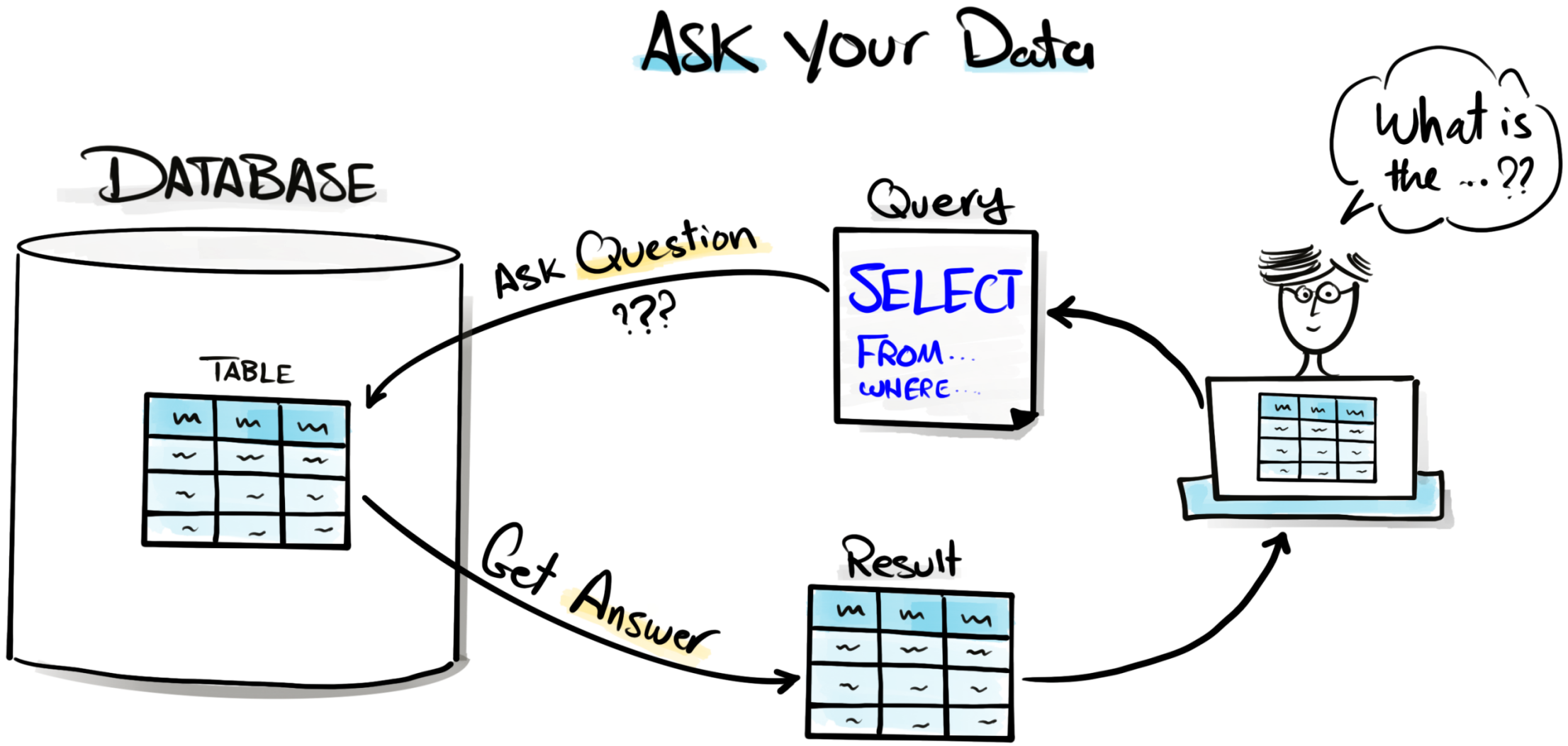
SELECT STATEMENT

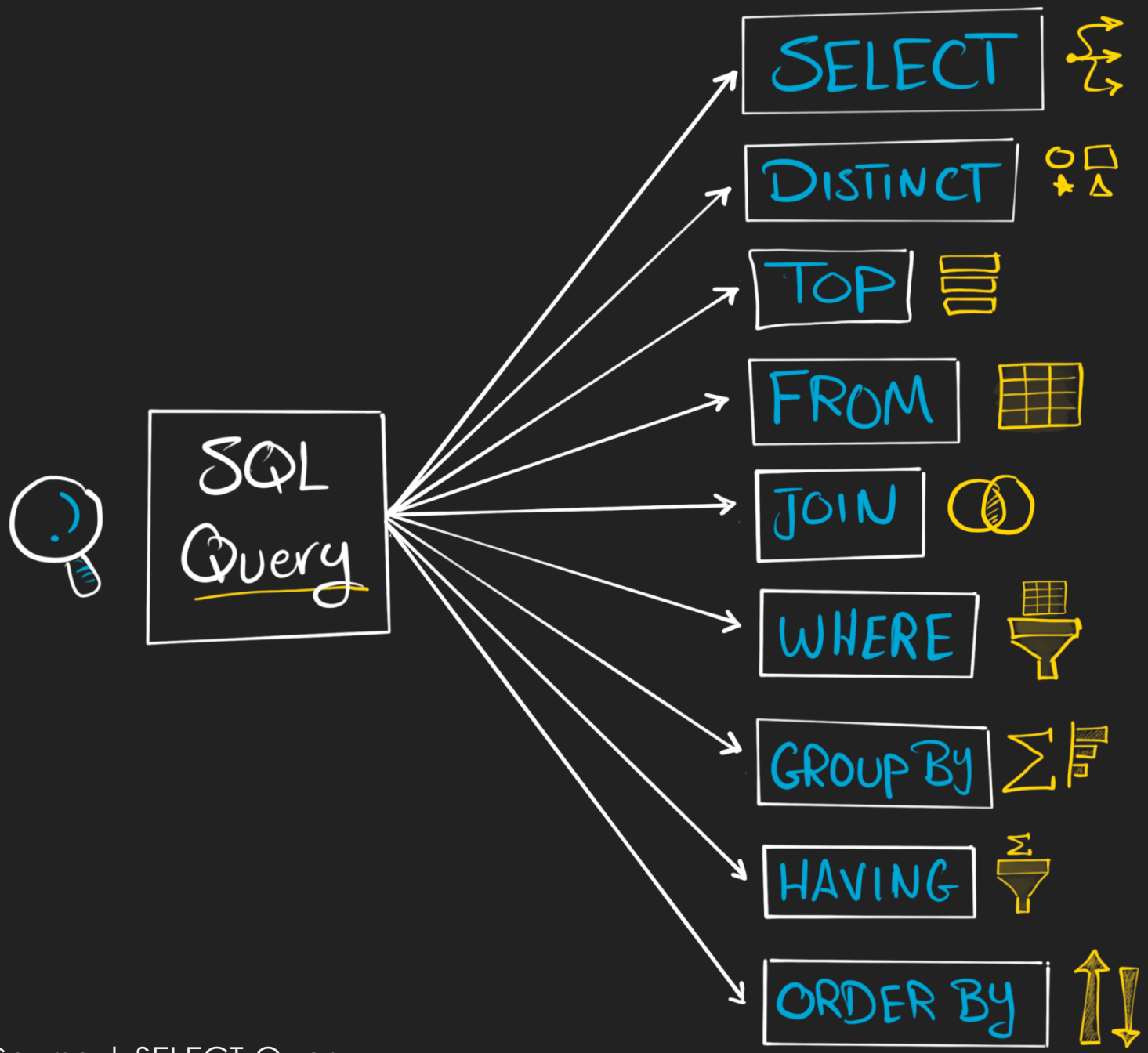
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SQL Course | SELECT Query




Types
SQL Commands
DQL
Data Query Language
SELECT
DDL
Data Definition Language
CREATE
DROP
ALTER
DML
Data Manipulation Language
INSERT
UPDATE
DELETE


Types
SQL Commands
DQL
Data Query Language
SELECT
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DROP
ALTER
DML
Data Manipulation Language
INSERT
UPDATE
DELETE


ASK your Data
DATABASE
TABLE
ASK Question
???
Query
SELECT
FROM...
WHERE...
What is the ...??
Get Answer
Result



DATA WITH BARAA

Data with Baraa

Database



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

SELECT *

FROM Table

Select * (All)

Retrieves All Columns (Everything)

From

Tells SQL Where to find Your Data

Database



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

2 SELECT *

1 FROM Table

Keep All Columns!!

id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

Database



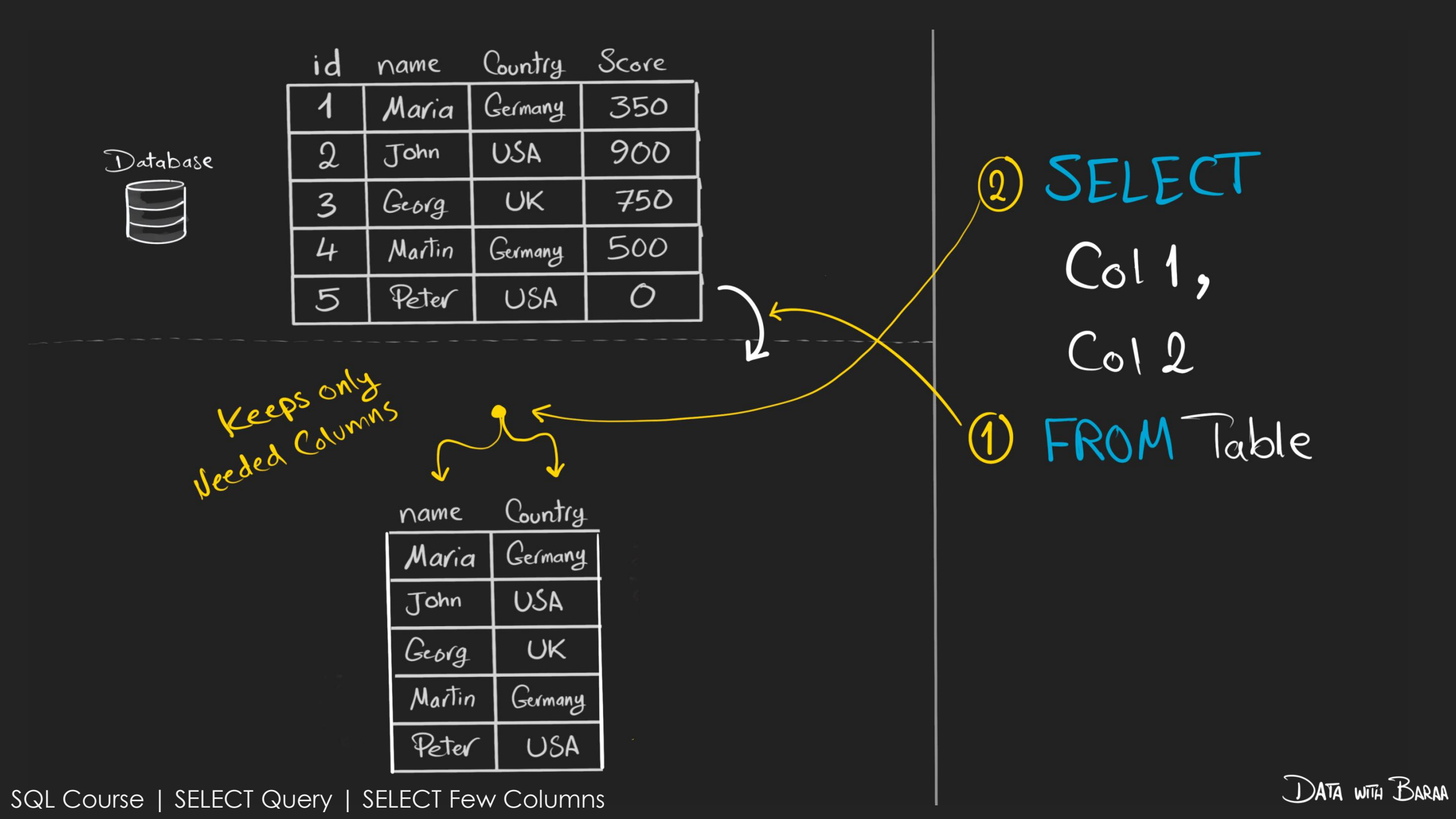
id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

```
SELECT
Col 1,
Col 2
FROM Table
```

Select Few Columns

Pick only the Columns You Need

instead of All.


id name Country Score
1 Maria Germany 350
2 John USA 900
3 Georg UK 750
4 Martin Germany 500
5 Peter USA 0
Database
② SELECT
Col 1,
Col 2
① FROM Table
Keeps only
Needed Columns
name Country
Maria Germany
John USA
Georg UK
Martin Germany
Peter USA

Database



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

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```
SELECT *
FROM Table
WHERE Condition
```

Where



Filters Your Data based on a Condition

Score Higher than 500

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Database



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

3 SELECT *

1 FROM Table

2 WHERE Condition

Score > 500



id	name	Country	Score	
1	Maria	Germany	350	✗
2	John	USA	900	✓
3	Georg	UK	750	✓
4	Martin	Germany	500	✗
5	Peter	USA	0	✗

Database



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

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```
SELECT *
FROM Table
ORDER BY Score DESC
```

Order By

Sort Your Data

- ASC
  - Lowest
  - Highest
- DESC
  - Highest
  - Lowest

Database



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

3 SELECT *

1 FROM Table

2 ORDER BY Score DESC

Highest



Lowest

id	name	Country	Score
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
1	Maria	Germany	350
5	Peter	USA	0

Database



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

Lowest

Highest

id	name	Country	Score
4	Martin	Germany	500
1	Maria	Germany	350
3	Georg	UK	750
2	John	USA	900
5	Peter	USA	0

Highest

Lowest

Highest

Lowest

③ SELECT *

① FROM Table

② ORDER BY

Country ASC,

Score DESC

DATA WITH BARAA

Database



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

Group By

Combines rows with the Same Value

Aggregates a Column By another Column

Total Score By Country

```
SELECT
    Country,      -- Category
    SUM(score)    -- Aggregation
FROM Table
GROUP BY Country
```

Database



	id	name	Country	Score
1	1	Maria	Germany	350
2	2	John	USA	900
3	3	Georg	UK	750
4	4	Martin	Germany	500
5	5	Peter	USA	0



1	Germany	850
2	USA	900
3	UK	750

```
(3) SELECT
    Country,
    SUM(score)
(1) FROM Table
(2) GROUP BY Country
```

DATA WITH BARAA

Database



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

Having

Filters Data After Aggregation

Can be used only with Group By

```
SELECT
    Country,
    SUM(score)
FROM Table
GROUP BY Country
HAVING SUM(score) > 800
```

Database



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0





Total Score > 800

4 SELECT
Country,
SUM(score)

1 FROM Table

2 GROUP BY Country

3 HAVING SUM(score) > 800

DATA WITH BARAA

Database



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0



Filter Your Data

① Before Aggregation

② After Aggregation

```
SELECT
  Country,
  SUM(score)
FROM Table
WHERE Score > 400
GROUP BY Country
HAVING SUM(score) > 800
```

Database



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

Distinct

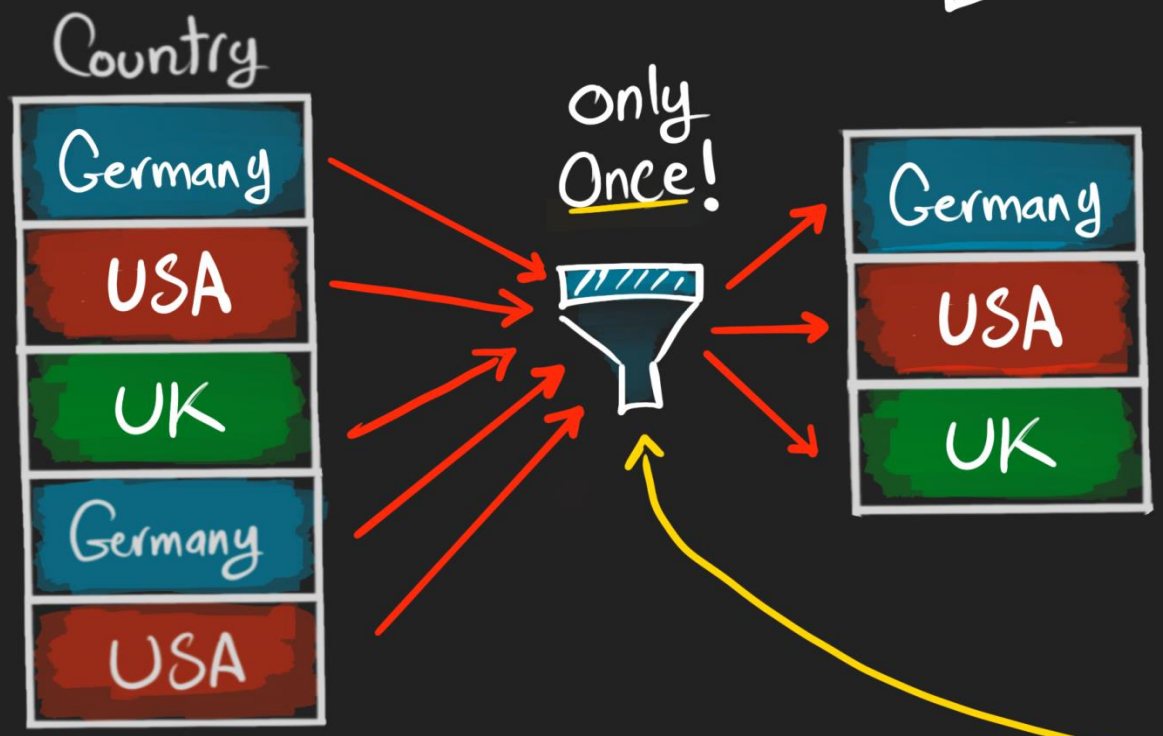
Removes Duplicates (Repeated Values)

Each Value appears only Once

```
SELECT DISTINCT
    Col
FROM Table
```

Database

id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0



3

② SELECT DISTINCT

Country

① FROM Table

Database



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

TOP (Limit)



Restrict the Number of Rows Returned

```
SELECT TOP 3
*
FROM Table
```

DATA WITH BARAA

Database



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

③ ② SELECT Top 3

\+

① FROM Table

	id	name	Country	Score
Row 1 →	1	Maria	Germany	350
Row 2 →	2	John	USA	900
Row 3 →	3	Georg	UK	750

Execute Order

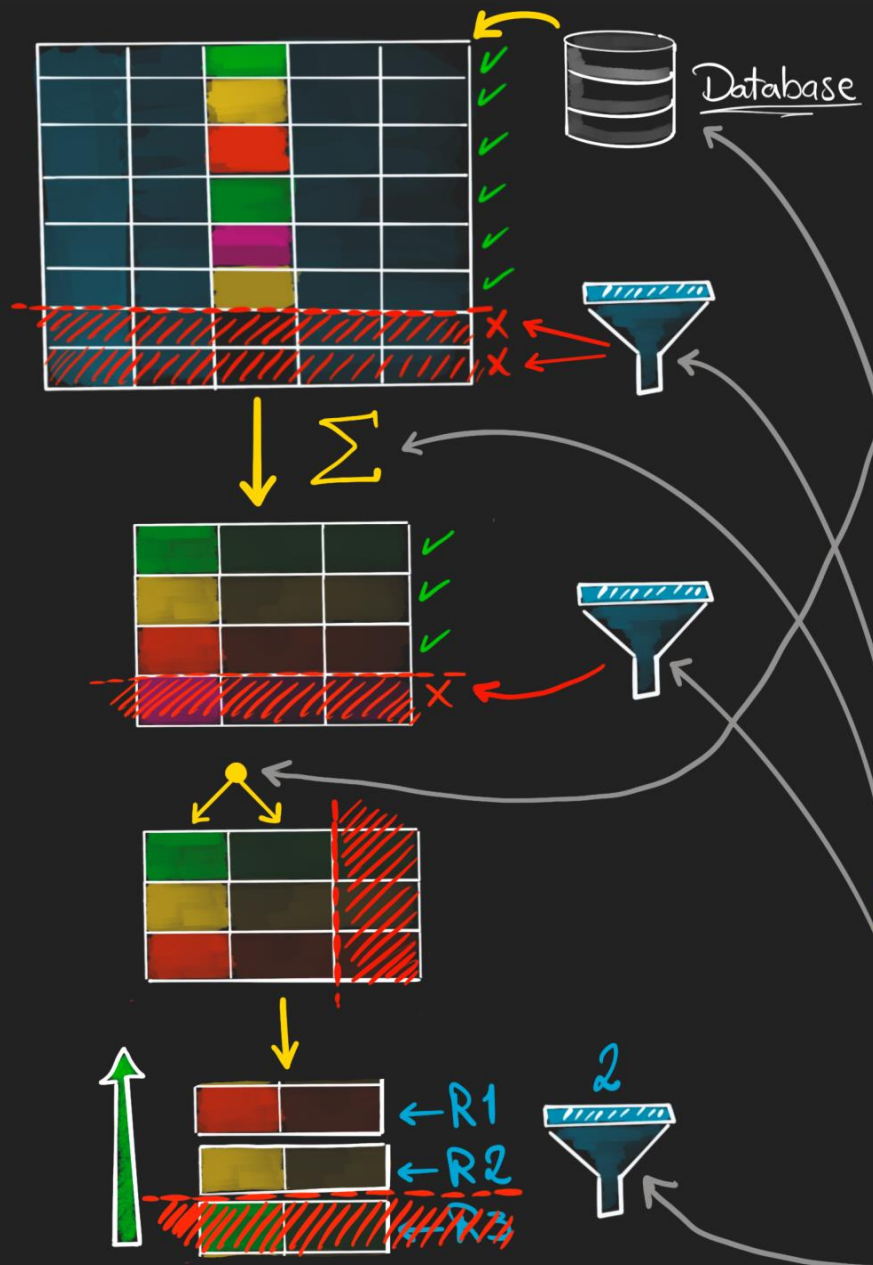


- Filter Columns → SELECT, DISTINCT, TOP 2
- Filter Duplicates → Col1,
- Filter Result Rows → SUM(Col2)
- FROM Table
- Filter Rows Before Aggregations → WHERE Col = 10
- GROUP BY Col1
- Filter Rows After Aggregations → HAVING SUM(Col2) > 30
- ORDER BY Col1 ASC

Coding Order

Execute Order

1. FROM
2. WHERE
3. GROUP BY
4. HAVING
5. SELECT DISTINCT
6. ORDER BY
7. TOP



Coding Order

```
5 SELECT DISTINCT Col1,
      SUM(Col2)
1 FROM Table
2 WHERE Col = 10
3 GROUP BY Col1
4 HAVING SUM(Col2) >30
6 ORDER BY Col1 ASC
  TOP 2
7
```

DATA WITH BARAA



DATA WITH BARAA

BONUS

Sketches

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SQL Course | SELECT Query



Database

Result

Execute Order

Customers

id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

① FROM

② SELECT *

id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

① FROM

Database

Customers

id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	Uk	750
4	Martin	Germany	500
5	Peter	USA	0



Result



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	Uk	750
4	Martin	Germany	500
5	Peter	USA	0

Execute Order

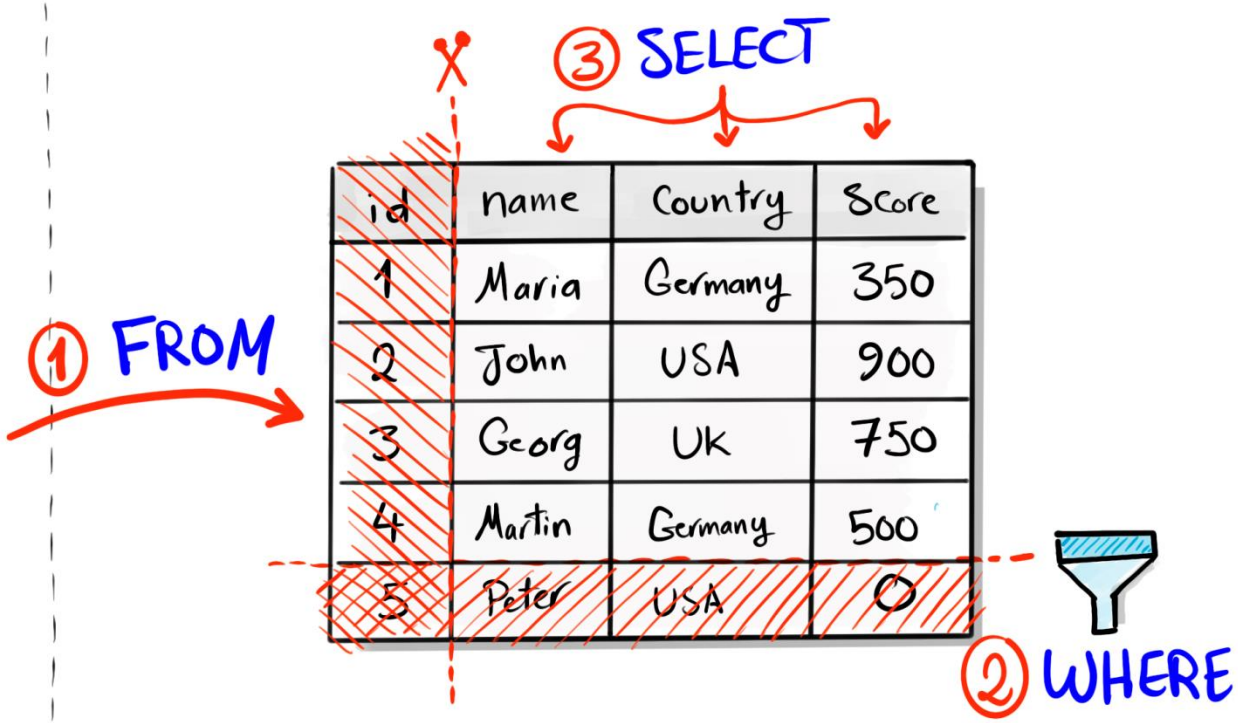
1 FROM

2 SELECT

Database

Customers

id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	Uk	750
4	Martin	Germany	500
5	Peter	USA	0



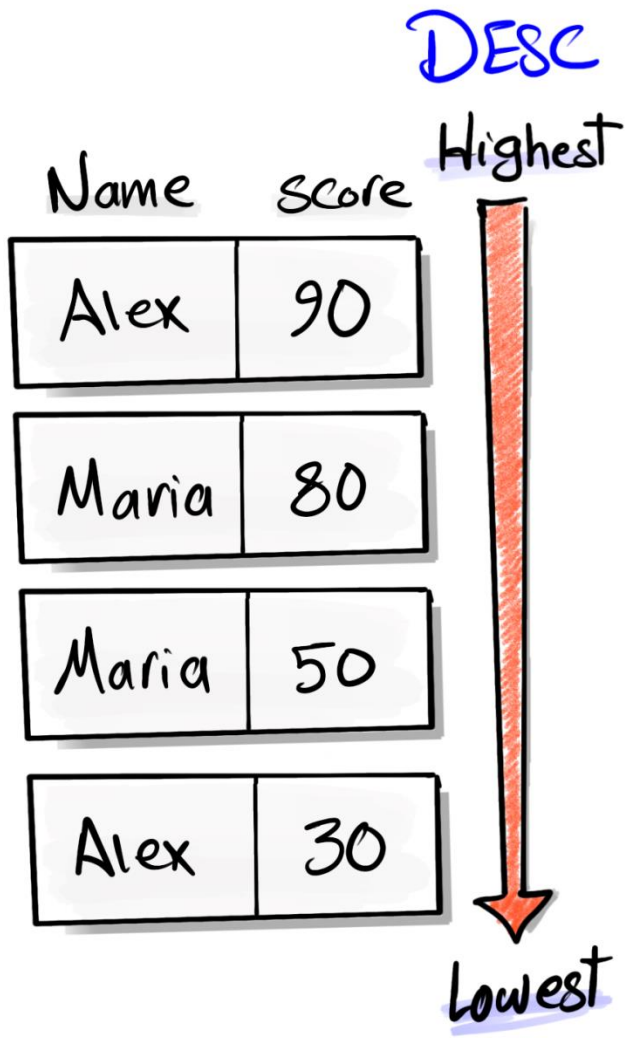
Result

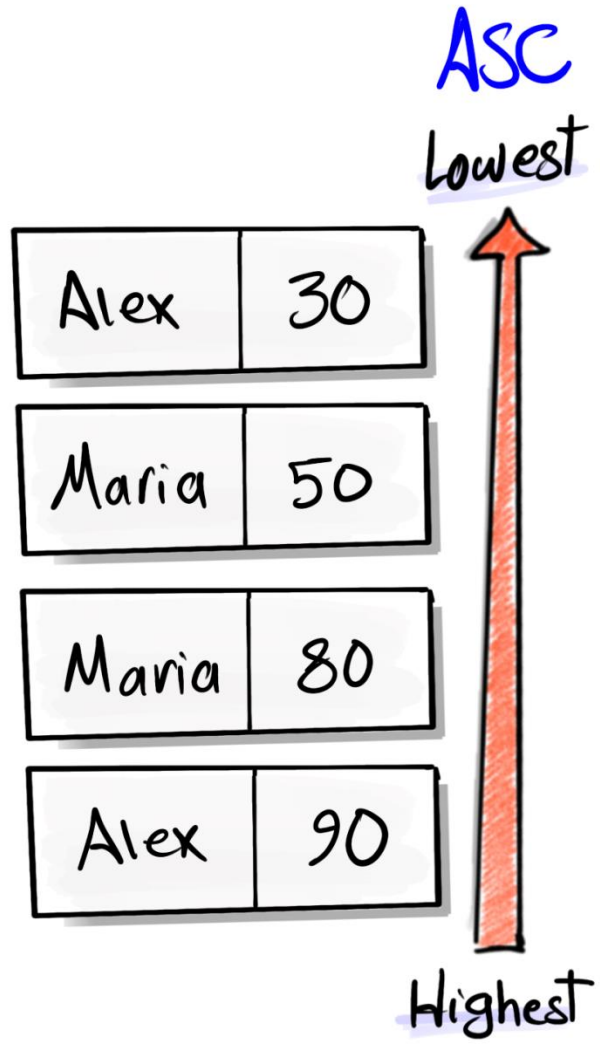
id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	Uk	750
4	Martin	Germany	500

Execute Order

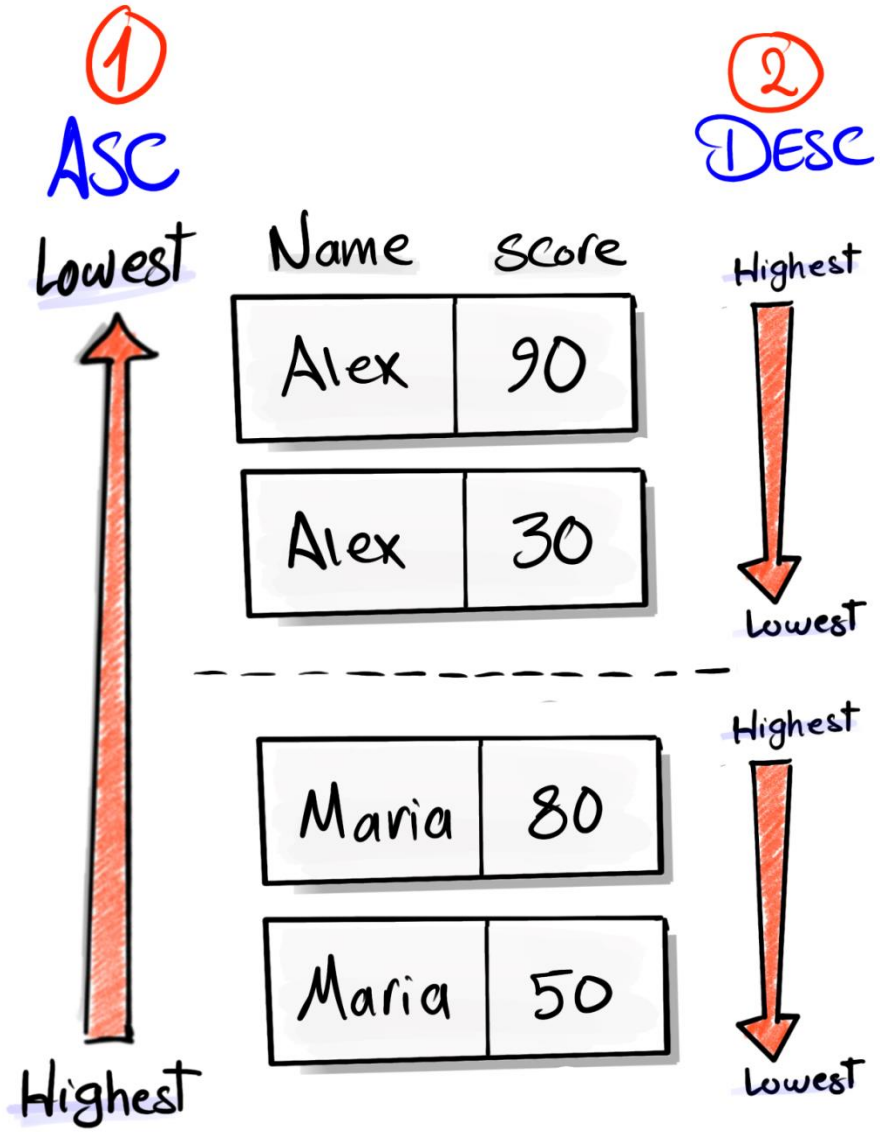
1. FROM
2. WHERE
3. SELECT

Name	score
Alex	90
Maria	50
Alex	30
Maria	80





Name	score
Alex	90
Maria	50
Alex	30
Maria	80



Database

Customers

id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	Uk	750
4	Martin	Germany	500
5	Peter	USA	0

Result

id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	Uk	750
4	Martin	Germany	500
5	Peter	USA	0

① FROM

② WHERE

③ GROUP BY

Country	Avg
Germany	425
USA	900
UK	750

④ SELECT

Execute Order

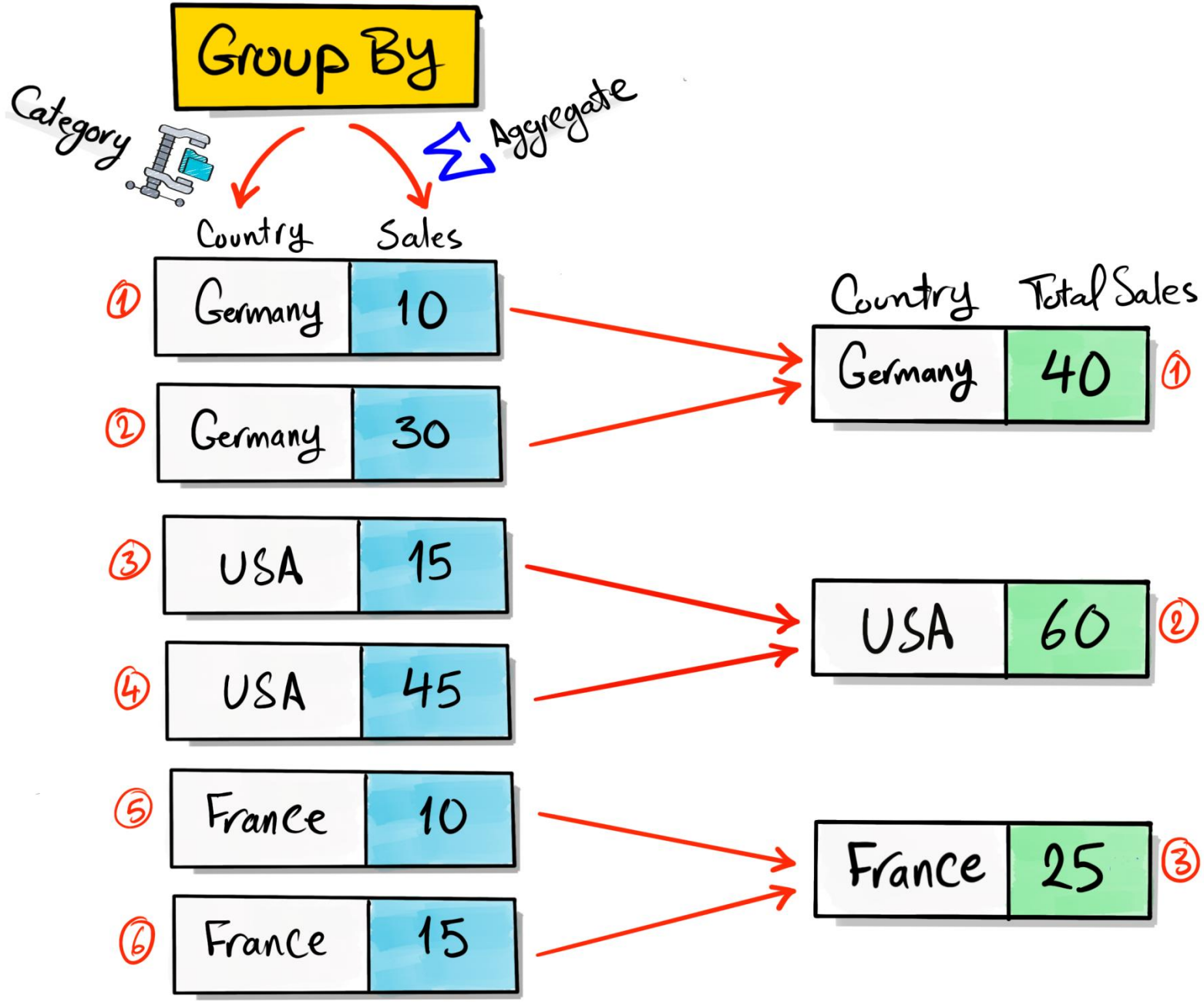
① FROM

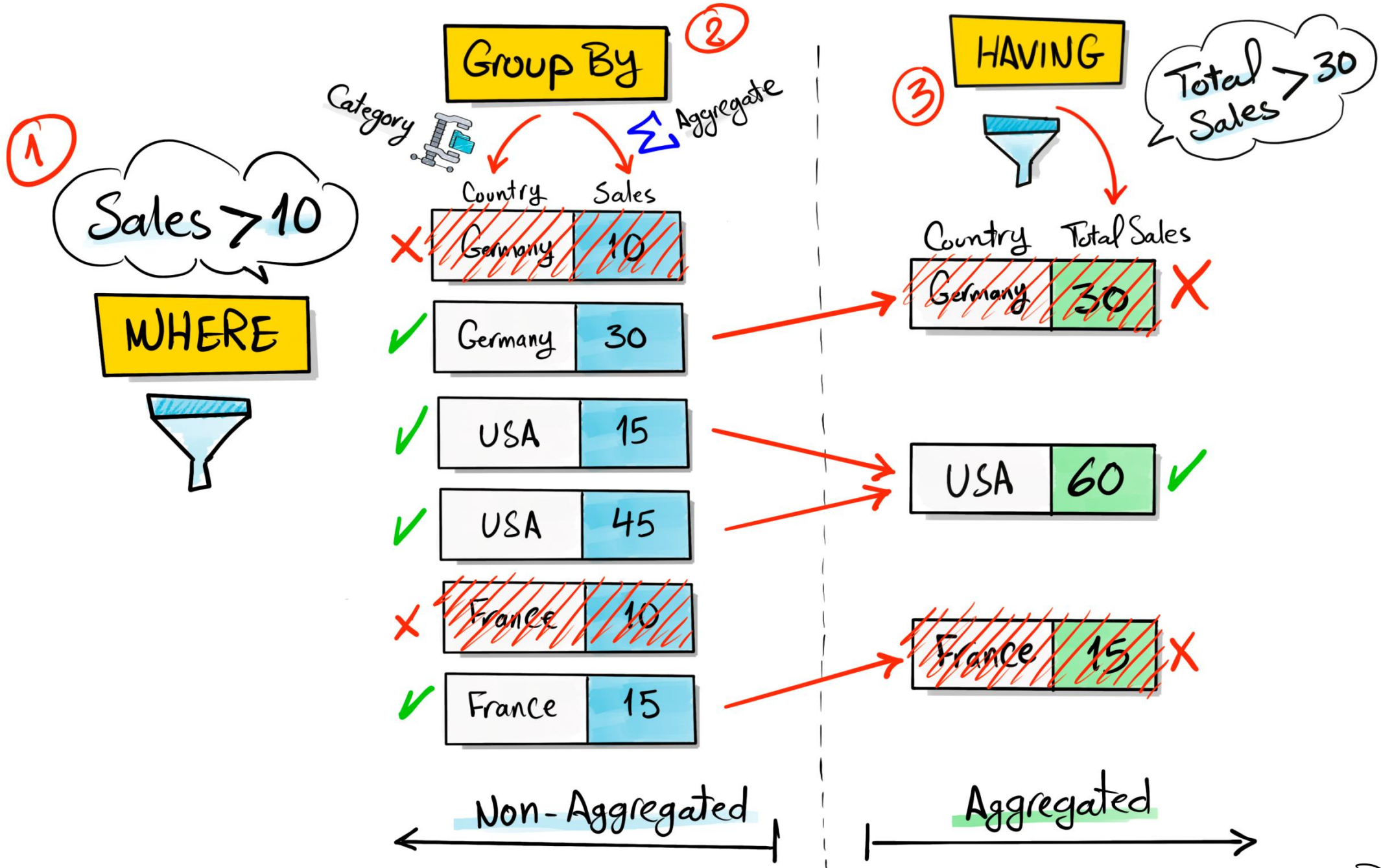
② WHERE

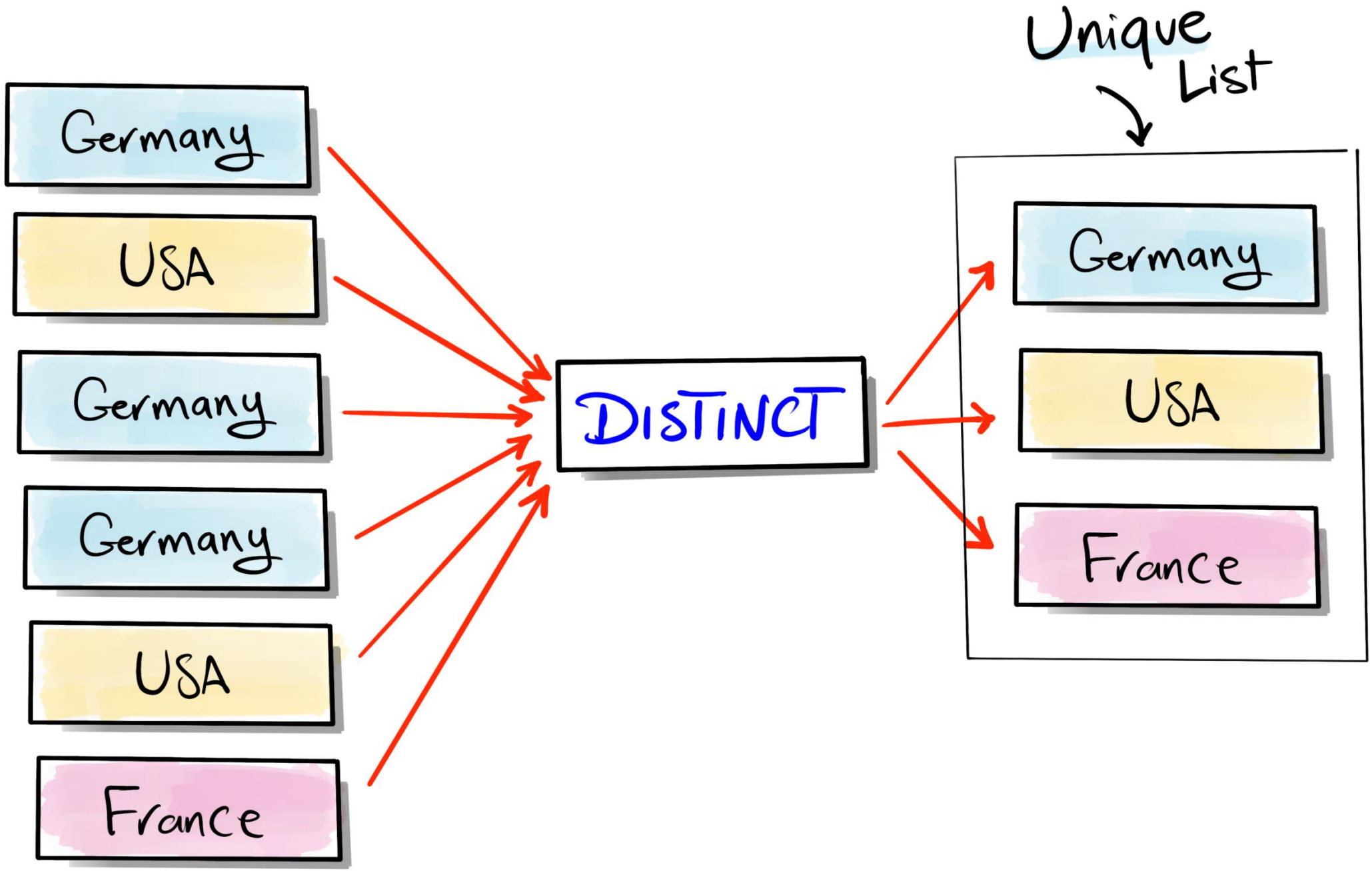
③ GROUP BY

④ SELECT

DATA WITH BARAA


Group By
Category
Aggregate
Country
Sales
1
Germany
10
2
Germany
30
3
USA
15
4
USA
45
5
France
10
6
France
15
Country
Total Sales
Germany
40
1
USA
60
2
France
25
3

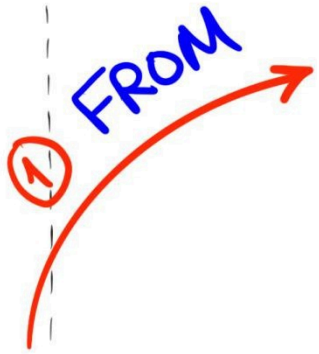

1
Sales > 10
WHERE
Group By
2
Category
Aggregate
Country
Sales
Germany
10
Germany
30
USA
15
USA
45
France
10
France
15
Non-Aggregated
3
HAVING
Total Sales > 30
Country
Total Sales
Germany
30
USA
60
France
15
Aggregated


Unique List
Germany
USA
Germany
Germany
USA
France
DISTINCT
Germany
USA
France

Database

Customers

id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	Uk	750
4	Martin	Germany	500
5	Peter	USA	0



Result

id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	Uk	750
4	Martin	Germany	500
5	Peter	USA	0

2 WHERE

Σ 3 GROUP BY

Country	Avg
Germany	425
USA	900
Uk	750

4 HAVING

5 SELECT

Country	Avg
Uk	750
USA	900

6 ORDER BY ASC

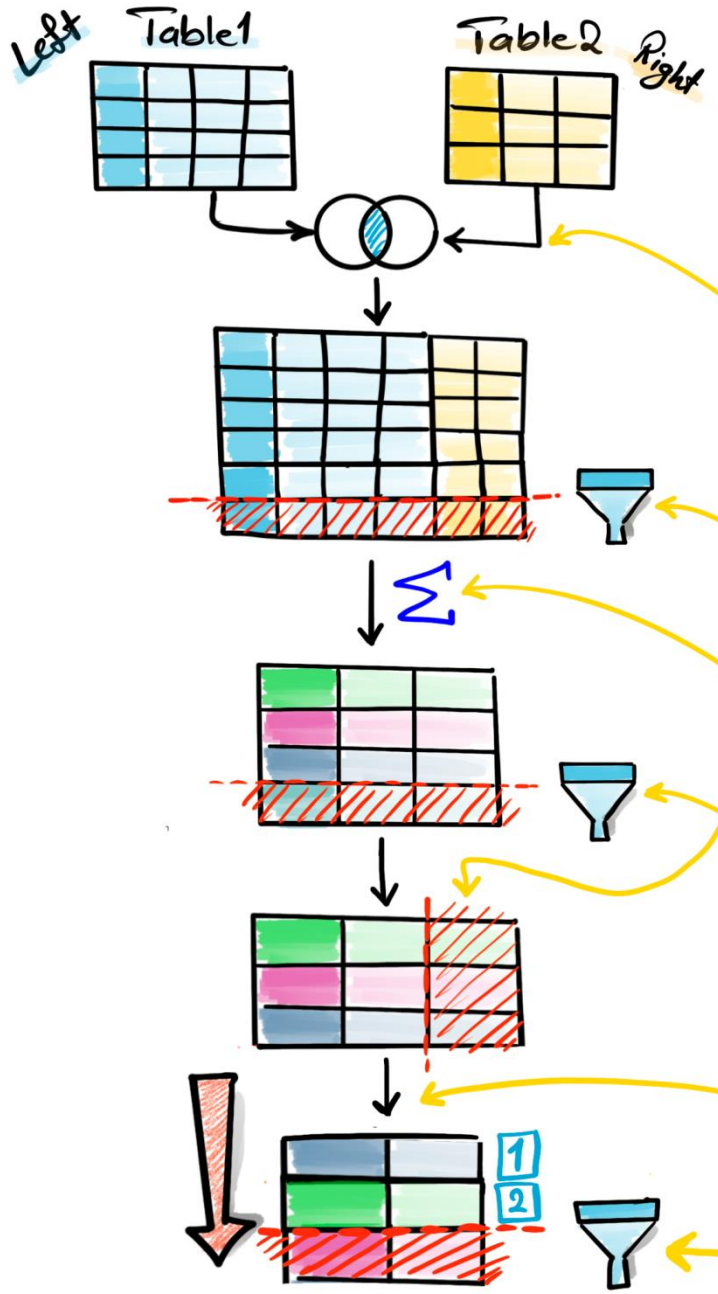
Execute Order

1. FROM
2. WHERE
3. GROUP BY
4. HAVING
5. SELECT
6. ORDER BY

DATA WITH BARAA

Execution Order

1. FROM JOIN
2. WHERE
3. GROUP BY
4. HAVING
5. SELECT DISTINCT
6. ORDER BY
7. TOP



Coding Order

```
SELECT DISTINCT                    -- 5
    a.Col,
    SUM(b.Col)
FROM Table1 AS a                   -- 1
JOIN Table2 AS b
    ON a.id = b.id
WHERE a.Col = 10                   -- 2
GROUP BY a.Col                     -- 3
HAVING SUM(b.Col)                  -- 4
ORDER BY a.Col                     -- 6
TOP 2                              -- 7
```